Hongyang Zhou

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EMPLOYMENT

Boston University — Boston, MA
Research Scientist — 2023/10-now

University of Helsinki — Helsinki, Finland
Postdoctoral Researcher — 2020/11-2023/05

University of Michigan — Ann Arbor, MI
Research Assistant — 2015/08-2020/09

EDUCATION

University of Michigan — Ann Arbor, MI
Ph.D of Climate and Space Sciences and Scientific Computing — 2015/08-2020/09

University of Science and Technology of China (USTC) — Anhui, China
Bachelor of Geophysics — 2011/09-2015/06

AWARDS

University of Michigan — 2020
Climate and Space Sciences Outstanding Doctoral Student Research Award

University of Michigan — 2016
Michigan Institute for Computational Discovery and Engineering Fellowship

USTC — 2015
Outstanding Teaching Scholarship

USTC — 2014
The First Prize, Outstanding Student Scholarship

EXPERIENCE

Planetary Magnetosphere Modeling — 2023/10-now
Supervisor: Chuanfei Dong
- Studying Earth and planetary magnetospheres with numerical simulations
- Developing kinetic shock and foreshock models

Hybrid Vlasov Ultra-Low Frequency (ULF) Wave Simulations — 2020/11-2023/05
Supervisor: Lucile Turc & Minna Palmroth
- Developed time-varying boundary conditions in Vlasiator and the postprocessing package Vlasiator.jl
- Studied ULF waves properties in Earth's magnetosheath and foreshock using global 2D simulations

SWMF Model Development — 2015-2020
Supervisor: Gábor Tóth
- Developed new boundary setups and extended MPI-OpenMP parallel code capability for BATS-R-US
- Developed an open-source package Batsrus.jl for efficient data processing, analysis and visualization
- Initiated a GPU-portable magnetohydrodynamic (MHD) solver with OpenACC, OpenMP & CUDA

MHD-EPIC Simulation of Ganymede's Magnetosphere — 2016-2020
Supervisor: Gábor Tóth & Xianzhe Jia
- Constructed a global magnetosphere model by coupling Hall MHD with Particle-In-Cell in generalized coordinates
- Analyzed the magnetic reconnection processes at Ganymede

TEACHING

Space Applications of Plasma Physics — University of Helsinki
Guest Lecturer — 2022

Plasma Physics — University of Helsinki
Guest Lecturer — 2022

Electrodynamics — USTC
Teaching Assistant — 2015

SKILLS

Programming Languages - Julia, Fortran, C, C++, MATLAB, Python, IDL, CUDA, Rust, Perl, LaTeX

3D Visualization & Rendering - Paraview, VisIt, Tecplot

Languages - English, Mandarin

PUBLICATIONS

Cross-Scale Coupling in Earth's Dayside Magnetosphere 2D Semi-Implicit PIC Simulation: From Global Geometry to Multi-Scale Kinetic Dynamics
Zhou, Hongyang *and Dong, Chuanfei and Chen, Yuxi and Zhang, Chi and Walsh, Brian M. and Zou, Ying and Li, Xinmin*, Journal of Geophysical Research: Space Physics, 2026

Planar Collisionless Shock Simulations with the Semi-implicit Particle-in-cell Model FLEKS
Zhou, Hongyang *and Chen, Yuxi and Dong, Chuanfei and Wang, Liang and Zou, Ying and Walsh, Brian M. and Tóth, Gábor*, The Astrophysical Journal Supplement Series, 2026

Properties and Radial Evolution of Solar Wind Turbulence Near Mercury's Orbit
Li, Xinmin and Dong, Chuanfei and Hadid, Lina Z. and Aizawa, Sae and Zhang, Chi and **Zhou, Hongyang** *and Wang, Liang and Gao, Jiawei and Slavin, James A.*, The Astrophysical Journal Letters, 2026

Direct Observations of Magnetic Reconnection in the Solar Wind Current Sheets near Mars
Zhang, Chi and Dong, Chuanfei and Li, Xinmin and Shen, Han-Wen and Halekas, Jasper and Phan, Tai and Mazelle, Christian and Harada, Yuki and **Zhou, Hongyang** *and Gao, Jiawei*, The Astrophysical Journal Letters, 2026

Physics-Informed Neural Networks for Modeling the Martian Induced Magnetosphere
Gao, Jiawei and Dong, Chuanfei and Zhang, Chi and Qin, Yilan and Shekarpaz, Simin and Li, Xinmin and Wang, Liang and **Zhou, Hongyang** *and Tadlock, Abigail*, Geophysical Research Letters, 2026

Global Mapping of Magnetic Field Topologies in the Martian Induced Magnetosphere
Zhang, Chi and Dong, Chuanfei and Xu, Shaosui and Halekas, Jasper and Luhmann, Janet and Shen, Han-Wen and **Zhou, Hongyang** *and Wang, Xiao-Dong and Gao, Jiawei and Li, Xinmin and others*, Journal of Geophysical Research: Space Physics, 2026

Spectral Properties and Energy Injection in Mercury's Magnetotail Current Sheet
Li, Xinmin and Dong, Chuanfei and Wang, Liang and Aizawa, Sae and Hadid, Lina Z. and Zhang, Chi and **Zhou, Hongyang** *and Slavin, James A. and Gao, Jiawei and Stumpo, Mirko and others*, Geophysical Research Letters, 2026

Revisiting Mars' Induced Magnetic Field and Clock Angle Departures Under Real-Time Upstream Solar Wind Conditions
Cheng, Zhihao and Zhang, Chi and Dong, Chuanfei and **Zhou, Hongyang** *and Gao, Jiawei and Tadlock, Abigail and Li, Xinmin and Wang, Liang*, Journal of Geophysical Research: Space Physics, 2025

Global Energy Transport and Conversion in the Solar Wind–Mars Interaction: MAVEN Observations
Zhang, Chi and Dong, Chuanfei and **Zhou, Hongyang** *and Halekas, Jasper and Li, Xinmin and Gao, Jiawei and Shen, Han-Wen and Wang, Xiao-Dong and Nilsson, Hans and Ramstad, Robin and others*, Journal of Geophysical Research: Planets, 2025

Observation of a Knotted Electron Diffusion Region in Earth's Magnetotail Reconnection
Li, Xinmin and Dong, Chuanfei and Ji, Hantao and Zhang, Chi and Wang, Liang and Giles, Barbara and **Zhou, Hongyang** *and Chen, Rui and Qi, Yi*, Geophysical Research Letters, 2025

Control of Solar Wind on Magnetic Field Fluctuations in the Subsolar Magnetosheath

Zou, Ying and Walsh, Brian M. and Chen, Yuxi and **Zhou, Hongyang** *and Raptis, Savvas*, Journal of Geophysical Research: Space Physics, 2025

The Early-Phase Growth of ULF Waves in the Ion Foreshock Observed in a Hybrid-Vlasov Simulation
Zhang, Kun and Dorfman, Seth and Turc, Lucile and Ganse, Urs and Shi, Chen and **Zhou, Hongyang** *and Palmroth, Minna*, Journal of Geophysical Research: Space Physics, 2025

BATSRUS GPU: Faster-than-real-time Magnetospheric Simulations with a Block-adaptive Grid Code
An, Yifu and Chen, Yuxi and **Zhou, Hongyang** *and Gaenko, A. and Tóth, Gábor*, The Astrophysical Journal, 2025

Interplay Between a Foreshock Bubble and a Hot Flow Anomaly Forming Along the Same Rotational Discontinuity
Turc, L. and Archer, M. O. and **Zhou, Hongyang** *and Pfau-Kempf, Y. and Suni, J. and Kajdič, P. and Blanco-Cano, X. and Dahani, S. and Battarbee, M. and Raptis, S. and others*, Geophysical Research Letters, 2025

Ion-Mediated Tearing and Kink Instabilities in the Earth's Magnetosphere: Hybrid-Vlasov Simulations
Zaitsev, I. and Cozzani, G. and Alho, M. and Horaites, K. and **Zhou, Hongyang** *and Kit, A. and Pfau-Kempf, Y. and Hoilijoki, S. and Ganse, U. and Battarbee, M. and others*, Journal of Geophysical Research: Space Physics, 2025

Role of ULF Waves in Reforming the Martian Bow Shock
Zhang, Chi and Dong, Chuanfei and Liu, Terry Z. and Mazelle, Christian and Raptis, Savvas and **Zhou, Hongyang** *and Fruchtman, Jacob and Halekas, Jasper and Li, Jing-Huan and Hanley, Kathleen G. and others*, AGU Advances, 2025

Anomalous Transient Enhancement of Planetary Ion Escape at Mars
Zhang, Chi and Dong, Chuanfei and **Zhou, Hongyang** *and Halekas, Jasper and Yamauchi, Masatoshi and Nilsson, Hans and Liu, Terry and Persson, Moa and Curry, Shannon and Dong, Yaxue and Futaana, Yoshifumi and Chen, Yuxi and Zhou, Muni and Suranga, Ruhunusiri and Hanley, Kathleen and Mazelle, Christian and Xu, Shaosui and Ramstad, Robin and Holmstrom, Mats and Chen, Li-Jen*, Nature Communications, 2025

Observational characteristics of electron distributions in the Martian induced magnetotail
Zhang, Chi and Dong, Chuanfei and **Zhou, Hongyang** *and Deca, Jan and Xu, Shaosui and Harada, Yuki and Curry, Shannon M and Mitchell, David L and Liu, Zhi-Yang and Qin, Junfeng and others*, Geophysical Research Letters, 2025

Interplay of magnetic reconnection and current sheet kink instability in the Earth's magnetotail
Cozzani, Giulia and Alho, M and Zaitsev, Ivan and **Zhou, Hongyang** *and Hoilijoki, Sanni and Turc, Lucile and Grandin, Maxime and Horaites, Konstantinos and Battarbee, Markus and Pfau-Kempf, Yann and others*, Geophysical Research Letters, 2025

The Vlasiator 5.2 Ionosphere–Coupling a magnetospheric hybrid-Vlasov simulation with a height-integrated ionosphere model
Ganse, Urs and Pfau-Kempf, Yann and Zhou, Hongyang and Juusola, Liisa and Workayehu, Abiyot and Kebede, Fasil and Papadakis, Konstantinos and Grandin, Maxime and Alho, Markku and Battarbee, Markus and others, Geoscientific Model Development Discussions, 2024

Hybrid-Vlasov modelling of ion velocity distribution functions associated with the Kelvin-Helmholtz instability with a density and temperature asymmetry
Tarvus, Vertti and Lurc, Lucile and **Zhou, Hongyang** *and Nakamura, Takuma and Settino, Adriana and Blasl, Kevin and Cozzani, Giulia and others*, AAS, 2024

Source of Drift-dispersed Electrons in Martian Crustal Magnetic Fields
Zhang, Chi and **Zhou, Hongyang** *and Dong, Chuanfei and Harada, Yuki and Yamauchi, Masatoshi and Xu, Shaosui and Nilson, Hans and Ebihara, Yusuke and Curry, Shannon M. and Qin, Junfeng and Mitchell, David L. and Brain, David A.*, Astrophysical Journal, 2024

Finding reconnection lines and flux rope axes via local coordinates in global ion-kinetic magnetospheric simulations

Alho, M., Cozzani, G., Zaitsev, I., Kebede, F. T., Ganse, U., Battarbee, M., Bussov, M., Dubart, M., Hoilijoki, S., Kotipalo, L., Papadakis, K., Pfau-Kempf, Y., Suni, J., Tarvus, V., Workayehu, A., **Zhou, H.***, and Palmroth, M.*, Annales Geophysicae, 2024

Kinetic signatures, dawn-dusk asymmetries, and flux transfer events associated with Mercury's dayside magnetopause reconnection from 3D MHD-AEPIC simulations
Li, Changkun and Jia, Xianzhe and Chen, Yuxi and Tóth, Gábor and **Zhou, Hongyang** *and Slavin, James A and Sun, Weijie and Poh, Gangkai*, JGR, 2024

Dayside Pc2 Waves Associated With Flux Transfer Events in a 3D Hybrid-Vlasov Simulation
Tesema, Facil and Palmroth, Minna and Turc, Lucile and **Zhou, Hongyang** *and Cozzani, Giulia and Alho, Markku and Pfau-Kempf, Yann and Horaites, Konstantinos and Zaitsev, Ivan and Grandin, Maxime*, GRL, 2024

Vlasiator. jl: A Julia package for processing Vlasiator data
Zhou, Hongyang, Journal of Open Source Software, 2023

FLEKS: A flexible particle-in-cell code for multi-scale plasma simulations
Chen, Yuxi and Tóth, Gábor and **Zhou, Hongyang** *and Wang, Xiantong*, CPC, 2023

Enabling technology for global 3D + 3V hybrid-Vlasov simulations of near-Earth space
Ganse, U. and Koskela, T. and Battarbee, M. and Pfau-Kempf, Y. and Papadakis, K. and Alho, M. and Bussov, M. and Cozzani, G. and Dubart, M. and George, H. and Gordeev, E. and Grandin, M. and Horaites, K. and Suni, J. and Tarvus, V. and Kebede, F. T. and Turc, L. and **Zhou, H.** *and Palmroth, M.*, Physics of Plasmas, 2023

Magnetospheric Response to a Pressure Pulse in a Three-Dimensional Hybrid-Vlasov Simulation
Horaites, Konstantinos and Rintamäki, E and Zaitsev, I and Turc, L and Grandin, M and Cozzani, G and **Zhou, Hongyang** *and Alho, M and Suni, J and Kebede, F and others*, JGR, 2023

Magnetotail plasma eruptions driven by magnetic reconnection and kinetic instabilities
Palmroth, Minna and Pulkkinen, Tuija I and Ganse, Urs and Pfau-Kempf, Yann and Koskela, Tuomas and Zaitsev, Ivan and Alho, Markku and Cozzani, Giulia and Turc, Lucile, **Zhou, Hongyang** *and others*, Nature Geoscience, 2023

First 3D hybrid-Vlasov global simulation of auroral proton precipitation and comparison with satellite observations
Grandin, M. and Luttikhuis, T. and Battarbee, M. and Cozzani, G. and **Zhou, H.** *and Turc, L. and Pfau-Kempf, Y.*, Journal of Space Weather and Space Climate, 2023

Magnetospheric responses to solar wind Pc5 density fluctuations: Results from 2D hybrid Vlasov simulation
Zhou, Hongyang *and Turc, Lucile and Pfau-Kempf, Yann and Battarbee, Markus and Tarvus, Vertti and Dubart, Maxime and George, Harriet and Cozzani, Giulia and Grandin, Maxime and Ganse, Urs*, Frontiers in Astronomy and Space Sciences, 2022

A global view of Pc3 wave activity in near-Earth space: Results from hybrid-Vlasov simulations
Turc, Lucile and **Zhou, Hongyang** *and Tarvus, Vertti and Ala-Lahti, Matti and Battarbee, Markus and Pfau-Kempf, Yann and Johlander, Andreas and Ganse, Urs and Dubart, Maxime and George, Harriet and others*, Frontiers in Astronomy and Space Sciences, 2022

Quasi-parallel Shock Reformation Seen by Magnetospheric Multiscale and Ion-kinetic Simulations
A. Johlander and M. Battarbee and L. Turc and U. Ganse and Y. Pfau-Kempf and M. Grandin and J. Suni and V. Tarvus and M. Bussov and **H. Zhou** *and others*, GRL, 2022

Estimating inner magnetospheric radial diffusion using a hybrid-Vlasov simulation
H. George and E. Kilpua and A. Osmane and S. Lejosne and L. Turc and M. Grandin and M. M. H. Kalliokoski and S. Hoilijoki and U. Ganse and M. Alho and M. Battarbee and M. Bussov and M. Dubart and A. Johlander and T. Manglayev and K. Papadakis and Y. Pfau-Kempf and J. Suni and **H. Zhou** *and M. Palmroth*, Frontiers in Astronomy and Space Sciences, 2022

Reconnection-driven dynamics at Ganymede's upstream magnetosphere: 3-D global Hall MHD and MHD-EPIC simulations
Zhou, Hongyang *and Tóth, Gábor and Jia, Xianzhe and Chen, Yuxi*, JGR, 2020

Efficient OpenMP parallelization to a complex MPI parallel magnetohydrodynamics code
Zhou, Hongyang *and Tóth, Gábor*, Journal of Parallel and Distributed Computing, 2020

Embedded kinetic simulation of Ganymede's magnetosphere: Improvements and inferences
Zhou, Hongyang *and Tóth, Gábor and Jia, Xianzhe and Chen, Yuxi and Markidis, Stefano*, JGR, 2019